

Hotel Room Booking & Reservation System:

1. Tech stack used (Backend / Frontend / DB / Payment) Backend: ASP.NET Core 8 Web API, C#, Entity Framework Core, JWT.

Frontend: React, Vite, JavaScript, Axios, React Router. **Database:** SQL Server.

Payment: Mock payment gateway flow.

2. Features fully completed :

User registration/login, JWT authentication, password hashing, role-based access, Admin/Manager/Guest workflows, hotel management, room management, availability checking, room holds, reservations, unique booking

references, mock payment and confirmation, manager approval, hotel activation/deactivation, booking oversight, payouts, reports, check-in/check-out, protected frontend routes.

3. Features partially completed :

Cancellation/refund workflow, advanced rate-plan and dynamic pricing, refresh-token handling, advanced reporting, audit logging, commission management and password reset.

4. Features not attempted (and

why) Production payment gateway, email notifications, advanced dispute management, reviews/ratings and complete automated load/concurrency testing were not implemented due to the

assignment timeline.

5. How double-booking was prevented under concurrency:

The system checks room availability using overlapping date-range validation before creating a hold/reservation. Active room holds and existing reservations are considered when checking availability. The booking flow is designed to prevent overlapping room bookings.

6. Known bugs / limitations :

Some advanced workflows such as complete refund/cancellation, automatic token refresh, advanced pricing and audit/reporting features require further development.

